

Basic Networking Skills for Digital Forensics

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2025/FWSD/11475

Fellowship in Web Application Security & Digital Forensics

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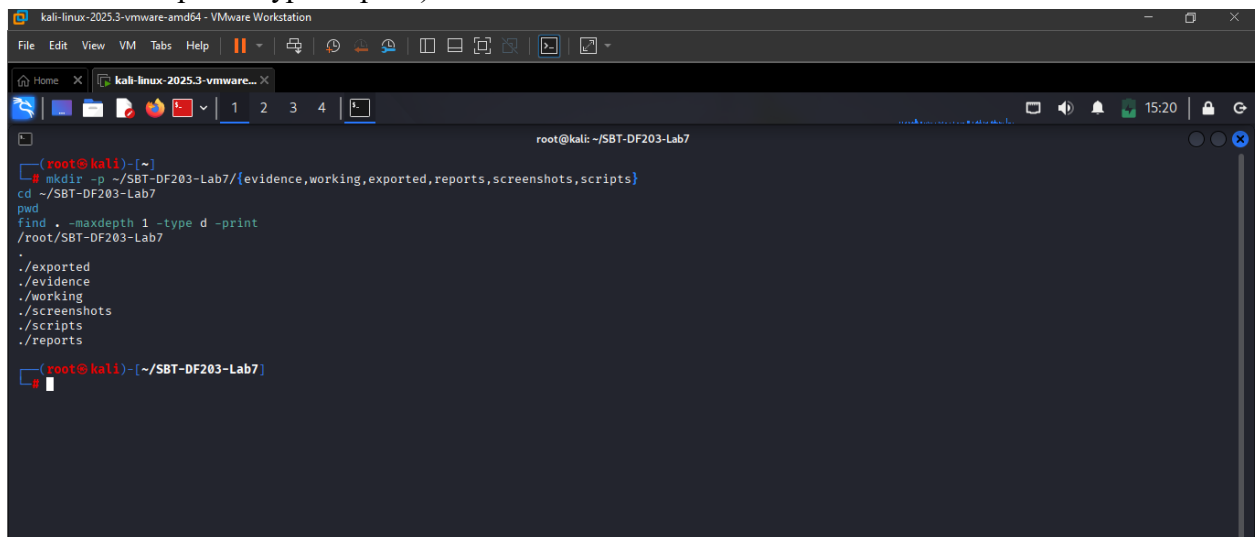
Lab Scenario

You are documenting normal DNS behaviour for an incident-response baseline. You will identify the configured resolver, capture controlled dig queries, analyze transaction IDs, names, record types, response codes, answers and TTL values, and correlate DNS resolution with a subsequent web or email connection.

Objective

To examine normal DNS behaviour by identifying the configured resolver, generating DNS queries using dig, capturing DNS traffic, analyzing query and response fields, and correlating DNS resolution with subsequent network connections.

1. Folder Structure Creation (mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
find . -maxdepth 1 -type d -print)



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
Home X kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab7
(root@kali)-[~]
# mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
/root/SBT-DF203-Lab7
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)-[~/SBT-DF203-Lab7]
```

2. Generating and downloading evidence (sudo apt update
sudo apt install -y dnsutils tshark wireshark)

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab7

(root@kali)~[~/SBT-DF203-Lab7]
# sudo apt update
# sudo apt install -y dnsutils tshark wireshark
Err:1 http://http.kali.org/kali kali-rolling InRelease
410 Gone [IP: 52.213.91.221 80]
Error: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/InRelease 410 Gone [IP: 52.213.91.221 80]
Error: The repository 'http://http.kali.org/kali kali-rolling InRelease' is no longer signed.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Notice: See apt-get(8) manpage for repository creation and user configuration details.
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  girl1.2-girepository-2.0 libcrypt-dev libjs-underscore libmbcrypto16 libraw23t64 openjdk-21-jre python3.13-venv
  imagemagick-7-common libgavl-1 libjxr0t64 libmpeg2encpp-2.1-0t64 libtsk19t64 openjdk-21-jre-headless rpcsvc-proto
  libann0 libgdal37 liblab-gamut1 libmpeg2encpp-2.1-0t64 libwireshark18 python3-fs samba-ad-dc
  libarmadillo14 libgirepository-1.0-1 liblqr-1-0 libplex2-2.1-0t64 libwiretap15 python3-gpg samba-ad-provision
  libavc1394-0 libgpgmepp6t64 liblqr-1-0 libpoppler147 libwmf-lite-0.2-7 python3.13 samba-dsdb-modules
  libblsc2-4 libpgm5 libmagiccore-7.q16-10 libportmidi0 libwsutil16 python3.13-dev python3.13-minimal
  libcdt5 libjs-jquery-ui libmagicwand-7.q16-10 libpython3.13-dev libzxing3 python3.13-minimal
Use 'sudo apt autoremove' to remove them.

Upgrading:
  bind9-dnsutils bind9-host bind9-libs

Summary:
  Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 2178
  Download size: 1,534 kB
  Space needed: 206 kB / 55.8 GB available

Err:1 http://http.kali.org/kali kali-rolling/main amd64 bind9-dnsutils amd64 1:9.20.27-2
410 Gone [IP: 54.72.47.234 80]

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

cat /etc/resolv.conf | tee reports/resolv_conf.txt

resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab7

(root@kali)~[~/SBT-DF203-Lab7]
# cat /etc/resolv.conf | tee reports/resolv_conf.txt
# Generated by NetworkManager
search localdomain
nameserver 192.168.126.2

(root@kali)~[~/SBT-DF203-Lab7]
# resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
Do you want to install it? (N/y)y
Reading package lists...
Building dependency tree...
Reading state information...
The following packages were automatically installed and are no longer required:
  girl1.2-girepository-2.0 libjxr0t64 libwmf-lite-0.2-7
  imagemagick-7-common liblab-gamut1 libwsutil16
  libann0 liblqr-1-0 libzxing3
  libarmadillo14 libmagiccore-7.q16-10 openjdk-21-jre
  libavc1394-0 libmagicwand-7.q16-10 openjdk-21-jre-headless
  libblsc2-4 libmbcrypto16 python3-fs
  libcdt5 libmpeg2encpp-2.1-0t64 python3-gpg
  libcrypt-dev libplex2-2.1-0t64 python3.13
  libgavl-1 libpoppler147 python3.13-dev
  libgdal37 libportmidi0 python3.13-minimal
  libgirepository-1.0-1 libpython3.13-dev python3.13-venv
  libgpgmepp6t64 libraw23t64 rpcsvc-proto
  libjs-jquery-ui libtsk19t64 samba-ad-dc
  libjs-underscore libwireshark18 samba-ad-provision
  libjxr-tools libwiretap15 samba-dsdb-modules
Use 'apt autoremove' to remove them.

Upgrading:
```

wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab7

(root@kali) ~/SBT-DF203-Lab7
# wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
--2026-09-22 15:47:34-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1804656 (1.7M) [application/octet-stream]
Saving to: 'dig_dns.pcap'

dig_dns.pcap          100%[=====] 1.72M  1.08MB/s  in 1.6s

2026-09-22 15:47:36 (1.08 MB/s) - 'dig_dns.pcap' saved [1804656/1804656]

(root@kali) ~/SBT-DF203-Lab7
#
```

Sha256sum Evidence

e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

evidence/dig_dns.pcap

e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

working/dig_dns_working.pcap

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
root@kali: ~/SBT-DF203-Lab7

(root@kali) ~/SBT-DF203-Lab7
# cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap

(root@kali) ~/SBT-DF203-Lab7
# sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 evidence/dig_dns.pcap
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 working/dig_dns_working.pcap

(root@kali) ~/SBT-DF203-Lab7
#
```

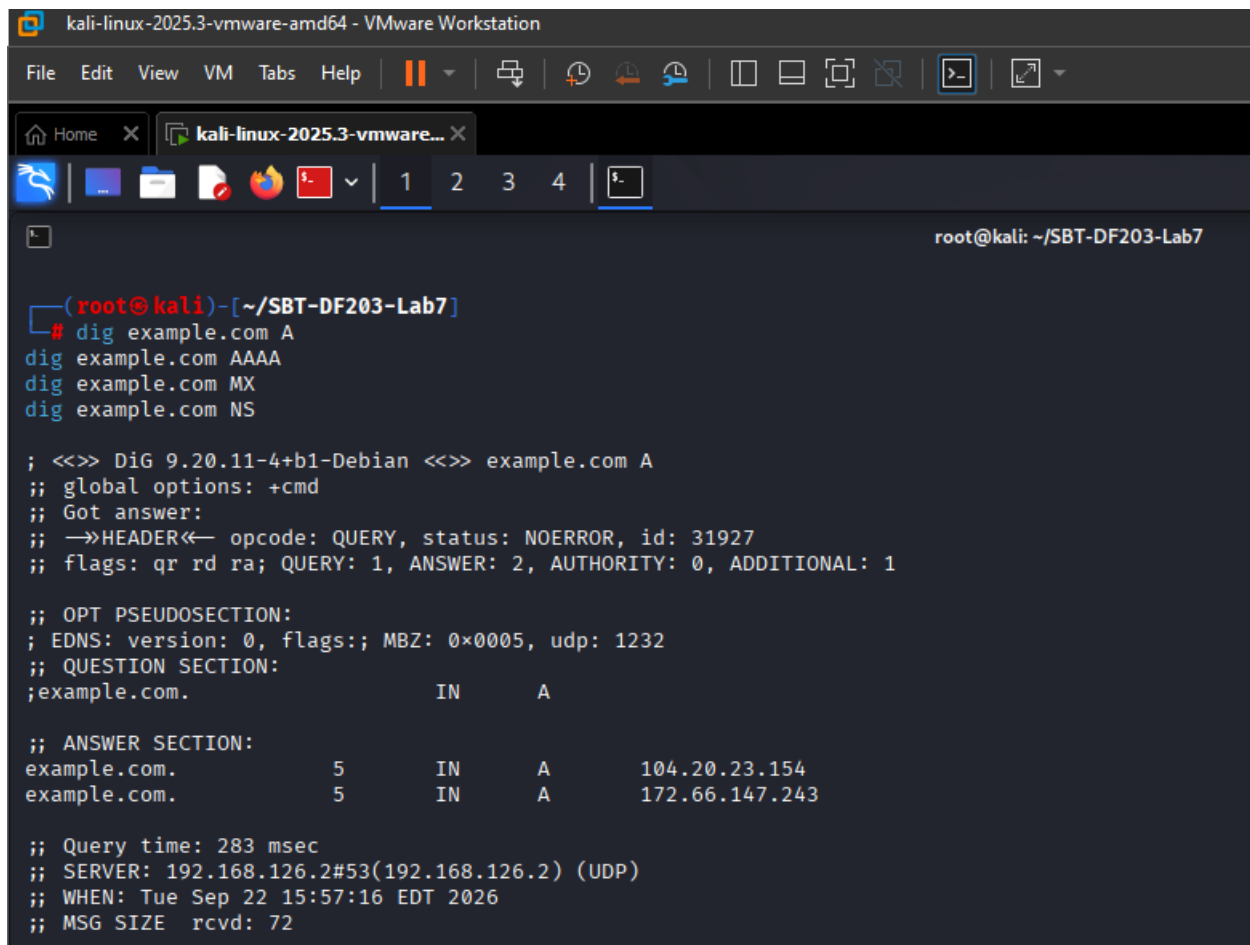
Mini Chain of Custody

Field	Entry
Case/lab identifier	SBT-DF203-Lab6
Trainee Name	Idriss Muhammad Abdullahi
Date and time started	20/09/2026 00.00
Evidence file names	Dns_cap

Source or generation method	Downloaded DNS capture and locally generated DNS traffic using dig
Original SHA-256	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
Working Copy SHA-256	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
Analysis WorkStation	Kali Linux
Notes on changes	Nil

Part A - Query DNS Records with dig

1. Dig example.com A



```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab7

(root@kali)-[~/SBT-DF203-Lab7]
# dig example.com A
dig example.com AAAA
dig example.com MX
dig example.com NS

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 31927
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 5       IN      A      104.20.23.154
example.com.                 5       IN      A      172.66.147.243

;; Query time: 283 msec
;; SERVER: 192.168.126.2#53(192.168.126.2) (UDP)
;; WHEN: Tue Sep 22 15:57:16 EDT 2026
;; MSG SIZE rcvd: 72

```

The Resolver in the SERVER line

192.168.126.2

The Response Information:

Flags: qr rd ra

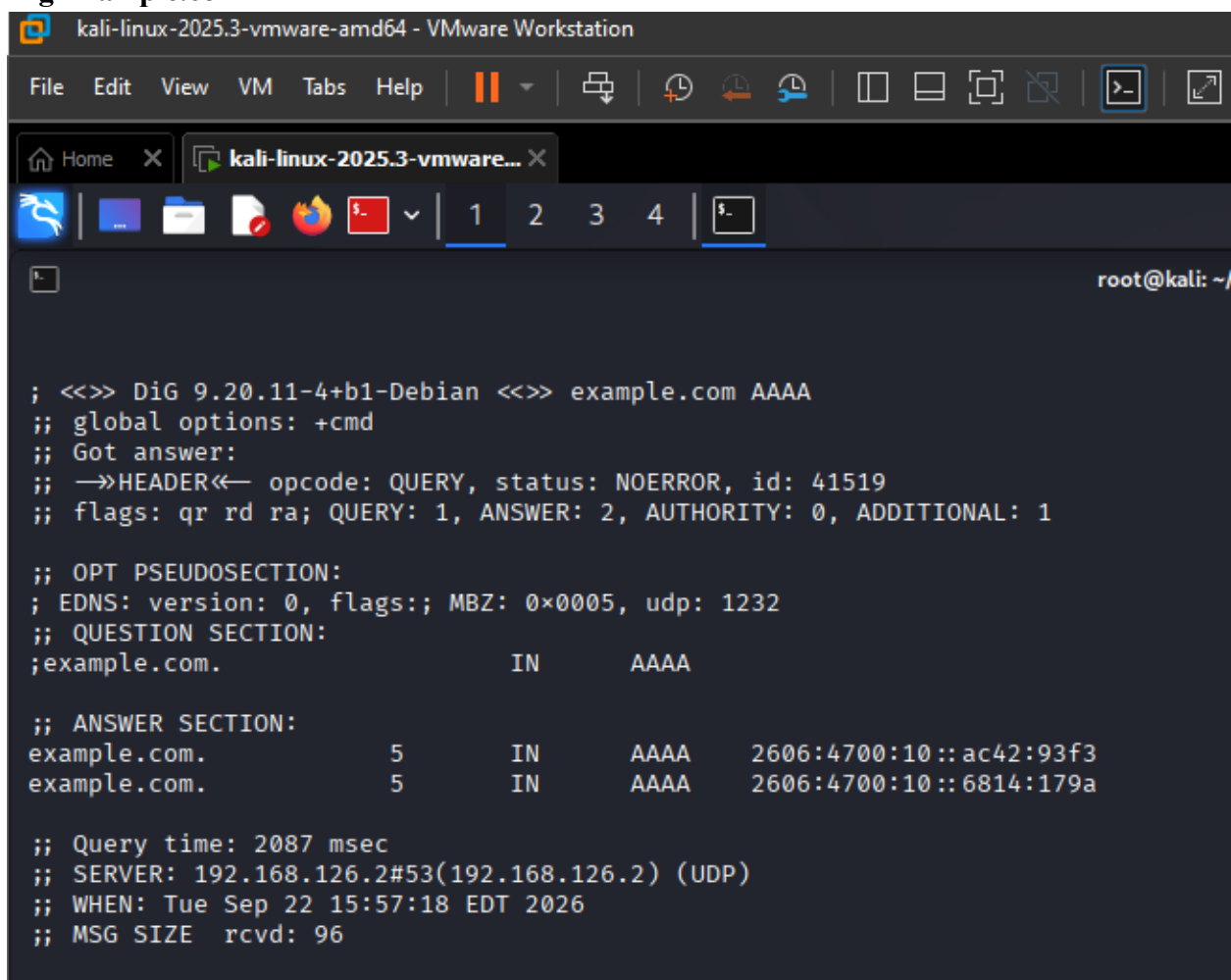
Status: NOERROR

Query: 1

Answer: 2

Query time: 283 msec

2. Dig Example.com AAAA



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware...
root@kali: ~/

; <>> DiG 9.20.11-4+b1-Debian <>> example.com AAAA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 41519
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      AAAA

;; ANSWER SECTION:
example.com.                  5       IN      AAAA    2606:4700:10::ac42:93f3
example.com.                  5       IN      AAAA    2606:4700:10::6814:179a

;; Query time: 2087 msec
;; SERVER: 192.168.126.2#53(192.168.126.2) (UDP)
;; WHEN: Tue Sep 22 15:57:18 EDT 2026
;; MSG SIZE rcvd: 96
```

The Resolver in the SERVER line

192.168.126.2

The Response Information:

Flags: qr rd ra

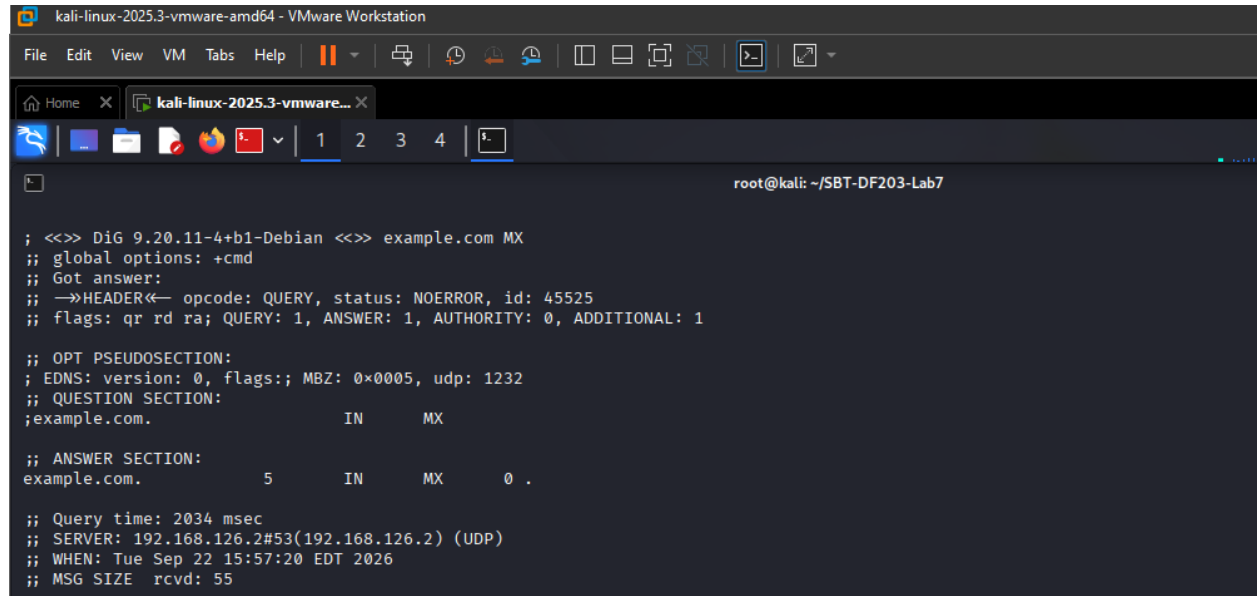
Status: NOERROR

Query: 1

Answer: 2

Query time: 2087 msec

3. Dig example.com MX



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64 X
root@kali: ~/SBT-DF203-Lab7

; <<>> DiG 9.20.11-4+b1-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 45525
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                5       IN      MX      0 .

;; Query time: 2034 msec
;; SERVER: 192.168.126.2#53(192.168.126.2) (UDP)
;; WHEN: Tue Sep 22 15:57:20 EDT 2026
;; MSG SIZE rcvd: 55
```

The Resolver in the SERVER line

192.168.126.2

The Response Information:

Flags: qr rd ra

Status: NOERROR

Query: 1

Answer: 1

Query time: 2034 msec

Dig example.com NS

```
; <<> DiG 9.20.11-4+b1-Debian <<> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 17797
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;example.com.                IN      NS

;; ANSWER SECTION:
example.com.                5       IN      NS      elliot.ns.cloudflare.com.
example.com.                5       IN      NS      hera.ns.cloudflare.com.

;; Query time: 2025 msec
;; SERVER: 192.168.126.2#53(192.168.126.2) (UDP)
;; WHEN: Tue Sep 22 15:57:22 EDT 2026
;; MSG SIZE rcvd: 95

(root@kali) [~/SBT-DF203-Lab7]
```

The Resolver in the SERVER line

192.168.126.2

The Response Information:

Flags: qr rd ra

Status: NOERROR

Query: 1

Answer: 2

Query time: 2025 msec

Meaning of DNS Record types

Record	Meaning
IN	Internet Class
A	IPv4 address record
AAAA	IPv6 address record
MX	Mail Exchange server record
NS	Names Server record

TTL's

TTL (Time To Live) of seconds indicates how long the DNS response may remain cached by resolvers. It does not represent the date the record was created.

Part B - Capture a Fresh DNS Query

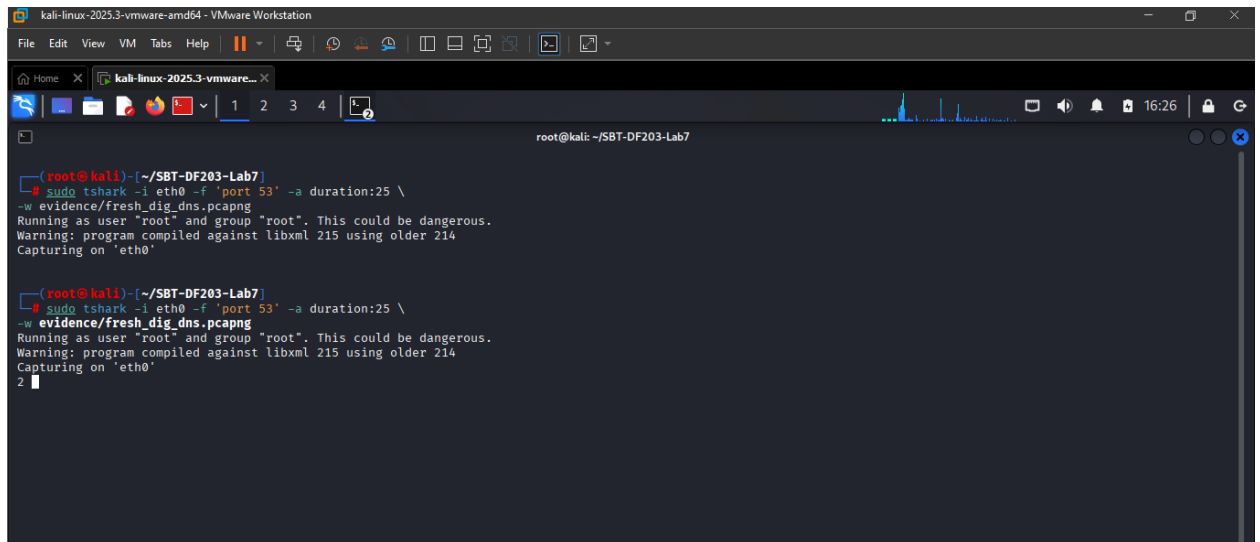
IFACE=eth0

sudo tshark -i "\$IFACE" -f 'port 53' -a duration:25 -w

evidence/fresh_dig_dns.pcapng &

sleep 3

dig +noedns example.com A >/dev/null



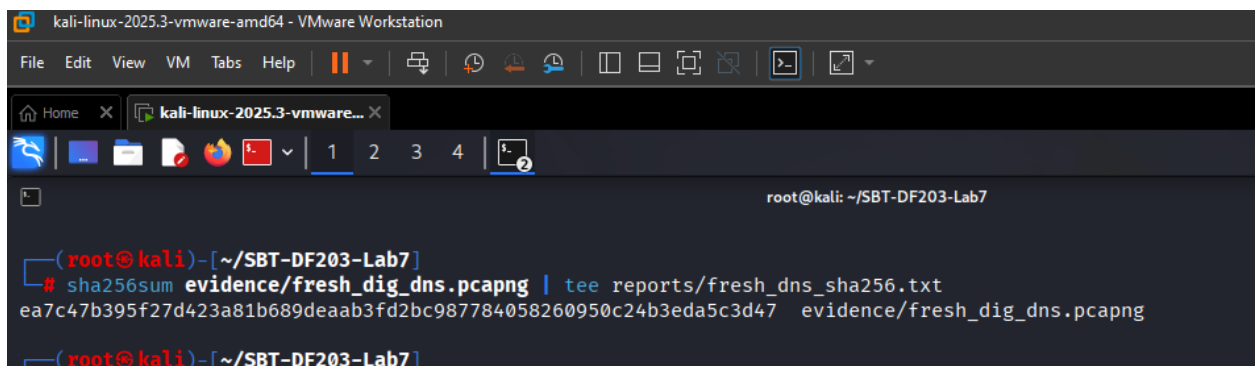
```
(root@kali)-[~/SBT-DF203-Lab7]
# sudo tshark -i eth0 -f 'port 53' -a duration:25 \
-w evidence/fresh_dig_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'

(root@kali)-[~/SBT-DF203-Lab7]
# sudo tshark -i eth0 -f 'port 53' -a duration:25 \
-w evidence/fresh_dig_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
2
```

SHA256sum

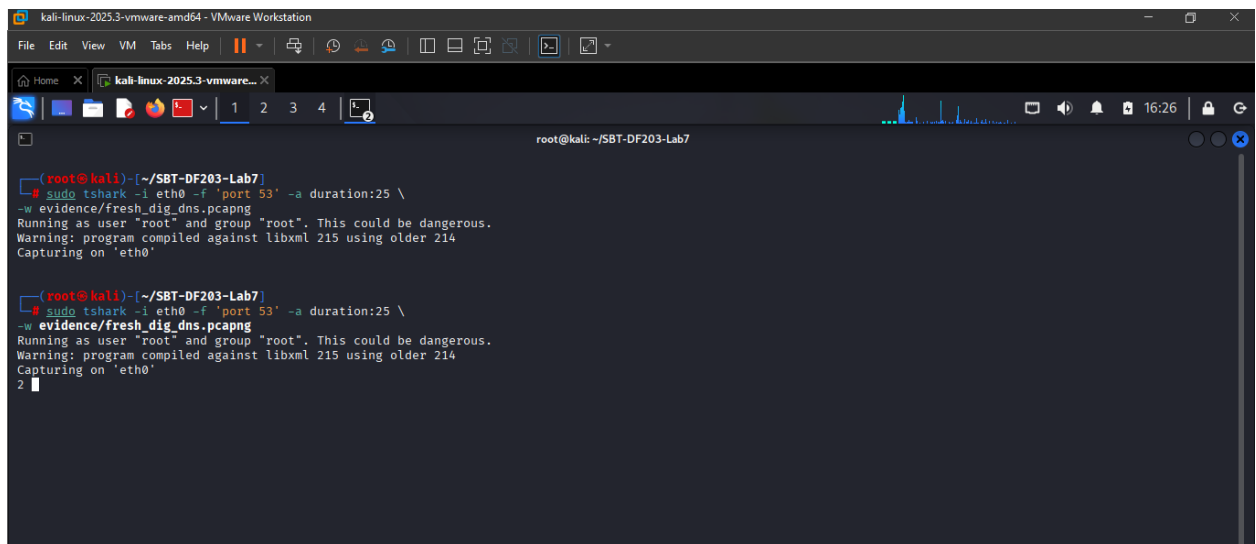
sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt

= ea7c47b395f27d423a81b689deaab3fd2bc987784058260950c24b3eda5c3d47



```
(root@kali)-[~/SBT-DF203-Lab7]
# sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
ea7c47b395f27d423a81b689deaab3fd2bc987784058260950c24b3eda5c3d47  evidence/fresh_dig_dns.pcapng

(root@kali)-[~/SBT-DF203-Lab7]
```



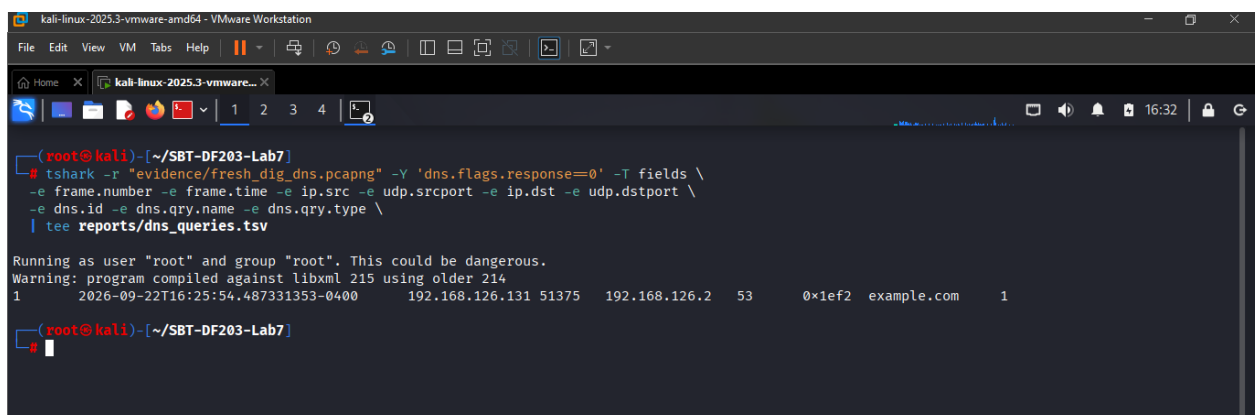
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab7

(root@kali)~/SBT-DF203-Lab7
# sudo tshark -i eth0 -f 'port 53' -a duration:25 \
-w evidence/fresh_dig_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'

(root@kali)~/SBT-DF203-Lab7
# sudo tshark -i eth0 -f 'port 53' -a duration:25 \
-w evidence/fresh_dig_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
2
```

A fresh DNS capture was generated using a controlled dig query for example.com. SHA-256 hashing was performed to verify evidential integrity.

**Part C - Extract DNS Query and Response Fields (tshark -r "evidence/fresh_dig_dns.pcapng" -Y 'dns.flags.response==0' -T fields **
**-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport **
**-e dns.id -e dns.qry.name -e dns.qry.type **
| tee reports/dns_queries.tsv)



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab7

(root@kali)~/SBT-DF203-Lab7
# tshark -r "evidence/fresh_dig_dns.pcapng" -Y 'dns.flags.response==0' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.qry.name -e dns.qry.type \
| tee reports/dns_queries.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 2026-09-22T16:25:54.487331353-0400 192.168.126.131 51375 192.168.126.2 53 0x1ef2 example.com 1

(root@kali)~/SBT-DF203-Lab7
#
```

**tshark -r "evidence/fresh_dig_dns.pcapng" -Y 'dns.flags.response==1' -T fields **
**-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport **

```
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
| tee reports/dns_responses.tsv
```

```
(root@kali) - [~/SBT-DF203-Lab7]
# tshark -r "evidence/fresh_dig_dns.pcapng" -Y 'dns.flags.response==1' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
| tee reports/dns_responses.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
2      2026-09-22T16:25:54.491128894-0400      192.168.126.2      53      192.168.126.131 51375      0x1ef2      0      2      172.66.147.243,104.20.23.154
5,5
(root@kali) - [~/SBT-DF203-Lab7]
```

Part D - Match Queries to Responses

Transaction ID	Query Name	Type	Client/Resolver	Response Code	Answer	TTL
0x1ef2	Example.com	1	192.168.126.131 51375	0	2	5,5

Calculating time

Query= 16:25:54.487331353

Response = 16:25:54.491128894

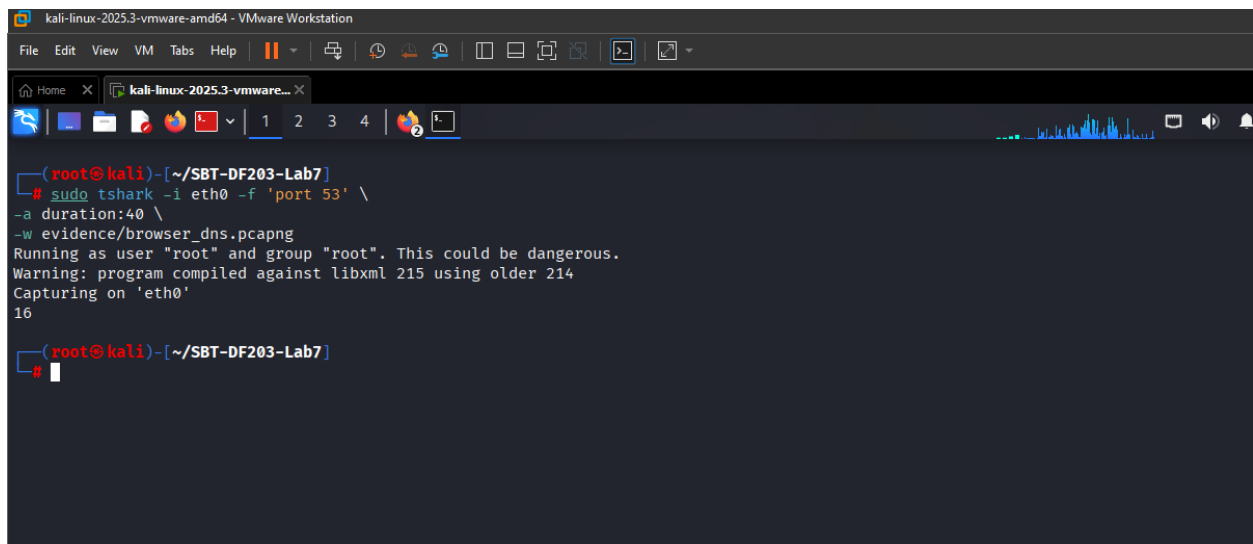
=0. 491128894 – 0. 487331353

= 0.003797541

Which is equals to 37.97541 msec

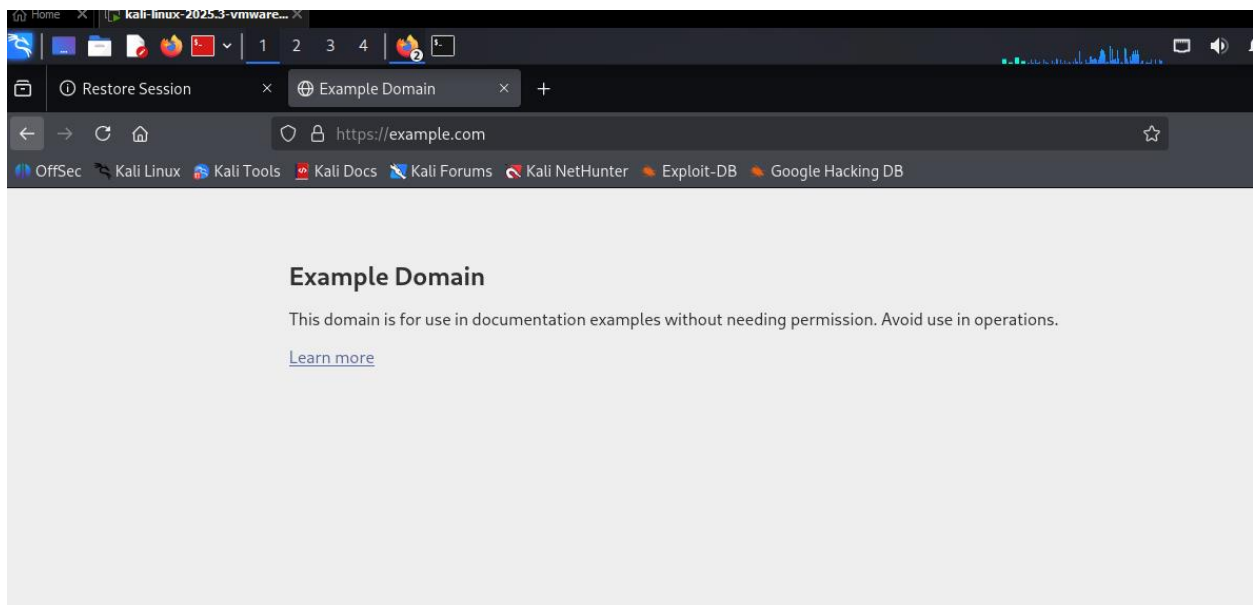
The DNS query used transaction ID 0x1ef2. The response originated from the queried resolver and contained a valid A record answer. Query-response latency was approximately 38 msec

Part E Browser DNS Analysis



```
(root@kali)-[~/SBT-DF203-Lab7]
# sudo tshark -i eth0 -f 'port 53' \
-a duration:40 \
-w evidence/browser_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
16

(root@kali)-[~/SBT-DF203-Lab7]
#
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
2

(root@kali)-[~/SBT-DF203-Lab7]
# sudo tshark -i eth0 -f 'port 53' \
-a duration:40 \
-w evidence/browser_dns.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'eth0'
16

(root@kali)-[~/SBT-DF203-Lab7]
# tshark -r evidence/browser_dns.pcapng \
-Y 'dns.flags.response==0' \
-T fields \
-e dns.qry.name \
-e dns.qry.type | sort | uniq -c | sort -nr
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
  2 example.org      1
  2 detectportal.firefox.com      1
  1 ipv4only.arpa    28
  1 ipv4only.arpa    1
  1 example.org      28
  1 detectportal.firefox.com      28

(root@kali)-[~/SBT-DF203-Lab7]
#
```

Browser activity generated DNS requests not only for the requested site but also for supporting resources including content delivery networks, fonts and cached services. Such behaviour is normal for web browsing.

Part F - Correlate DNS with Subsequent Connections

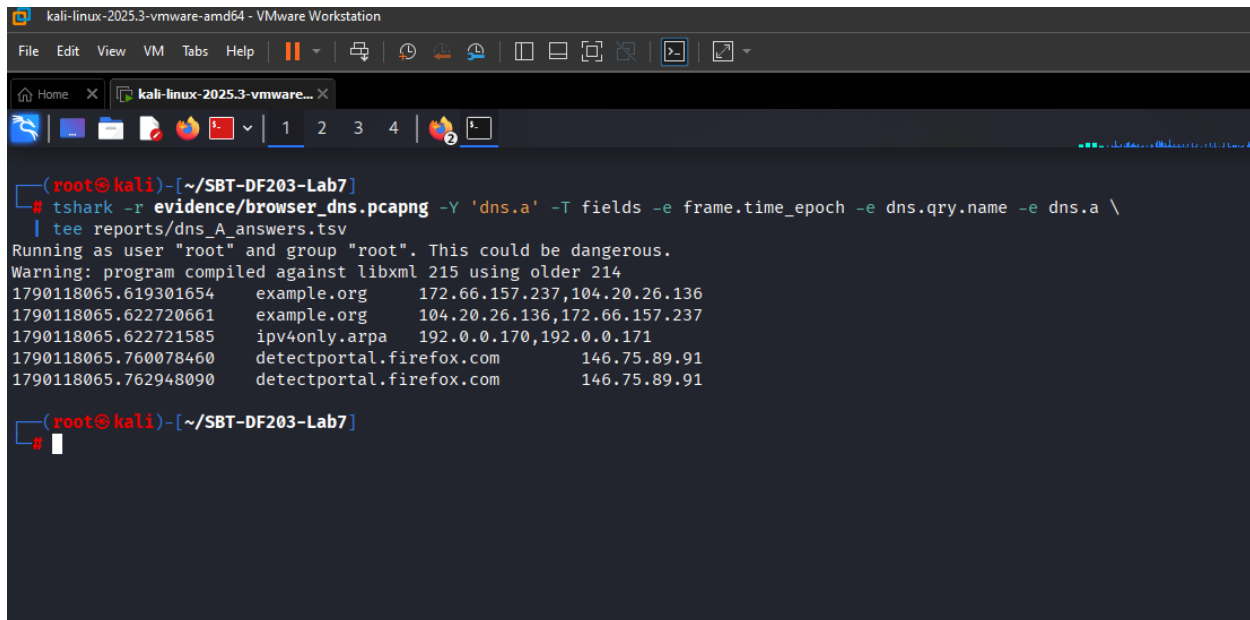
Extracting DNS A answers

```
tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv
```

Ans.

```
1790118065.619301654 example.org 172.66.157.237,104.20.26.136
1790118065.622720661 example.org 104.20.26.136,172.66.157.237
```

1790118065.622721585	ipv4only.arpa	192.0.0.170,192.0.0.171
1790118065.760078460	detectportal.firefox.com	146.75.89.91
1790118065.762948090	detectportal.firefox.com	146.75.89.91



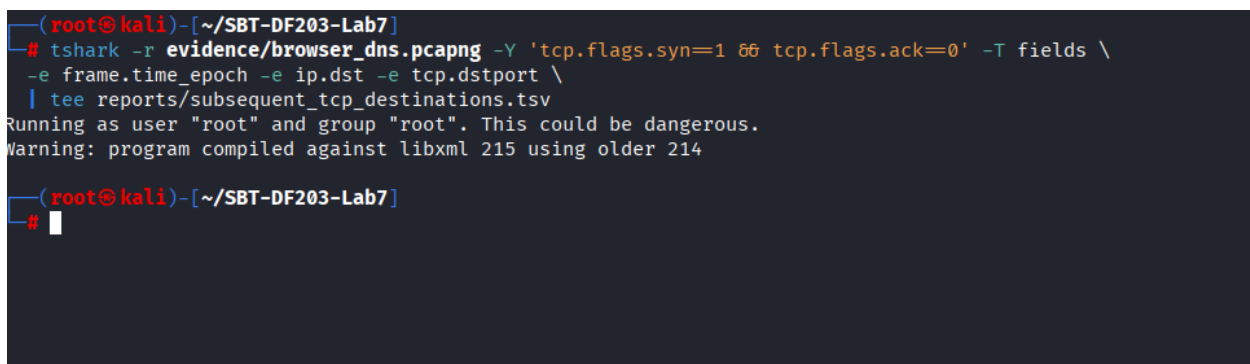
```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
1 2 3 4
(root@kali)~[~/SBT-DF203-Lab7]
# tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1790118065.619301654 example.org 172.66.157.237,104.20.26.136
1790118065.622720661 example.org 104.20.26.136,172.66.157.237
1790118065.622721585 ipv4only.arpa 192.0.0.170,192.0.0.171
1790118065.760078460 detectportal.firefox.com 146.75.89.91
1790118065.762948090 detectportal.firefox.com 146.75.89.91
(root@kali)~[~/SBT-DF203-Lab7]
#

```

DNS resolution returned IP address 172.66.157.237 for example.com. Subsequently, the client initiated a TCP connection to the same IP on port 443. This indicates that the DNS response was used to establish the HTTPS session.

Extracting later TCP destinations



```

(root@kali)~[~/SBT-DF203-Lab7]
# tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e frame.time_epoch -e ip.dst -e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
(root@kali)~[~/SBT-DF203-Lab7]
#

```

Conclusions

The captured DNS traffic exhibited normal resolution behaviour. Queries were successfully matched to responses using transaction IDs, appropriate answer records and TTL values

were returned, and resolved addresses were observed being used for subsequent network connections. No anomalous or suspicious DNS activity was identified during the analysis.